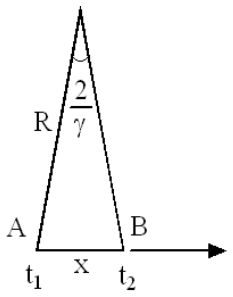
3：

P

$2/\gamma$



$$x = \frac{2R}{\gamma} . \qquad (1)$$

$\beta = v / c$

$$t_2 - t_1 = \frac{2R}{\gamma c \beta} . \qquad (2)$$

A B

$$t' = \frac{2R}{\gamma c} , \qquad (3)$$

$$\Delta T = t_2 - t_1 - t' . \qquad (4)$$

$$\gamma = \frac{mc^2}{m_0 c^2} = \frac{1}{\sqrt{1-\beta^2}} , \qquad (5)$$

$1-\beta \to 0$

$$\frac{2}{\beta}(1-\beta) = \frac{1}{\gamma^2} . \qquad (6)$$

(4)

$$\Delta T = \frac{R}{c\gamma^3} . \qquad (7)$$

$$\Delta T = 9.51 \times 10^{-20}\text{s} \qquad (8)$$